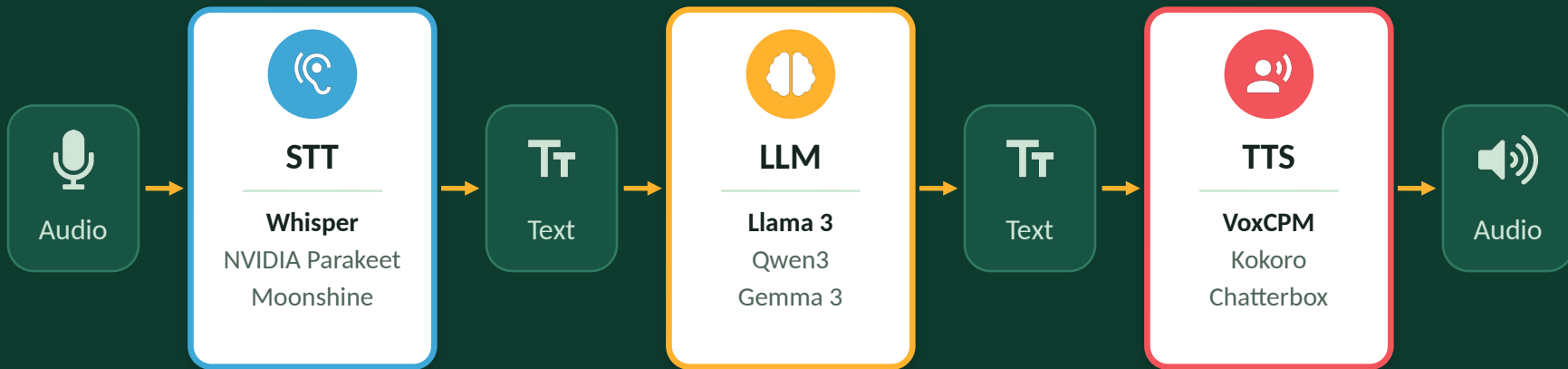


The Cascaded Pipeline

Three specialist models, chained through text — the “Unix philosophy” of voice AI.



Bold = reference stack · others are open-source drop-in alternatives

Key idea: text is the contract between every stage — so each model can be tuned, swapped, inspected and scaled on its own.

Real-Time Plumbing

What turns three models into a conversation you can actually talk to.



Transport

WebRTC for browser & mobile: UDP, Opus, jitter buffer, echo cancellation. WebSocket for servers; SIP for phones.

OPEN-SOURCE TOOLS

LiveKit
mediasoup
Janus Gateway



VAD

Labels each ~30 ms frame as speech or silence. Gates STT and fires barge-in when the user talks over the bot.

OPEN-SOURCE TOOLS

Silero VAD
WebRTC VAD
TEN VAD



Turn Detection

Decides whether the user has finished or is only pausing. Semantic models beat fixed silence timeouts.

OPEN-SOURCE TOOLS

LiveKit Turn Detector
Pipecat Smart Turn
TEN Turn Detection



Orchestration

Streams audio and text frames between stages; handles interruptions, TTS cancellation and conversation context.

OPEN-SOURCE TOOLS

Pipecat
LiveKit Agents
TEN Framework

Why Cascaded Works

Text in the middle buys control, clarity and flexibility.



Best-in-class reasoning & RAG

Text LLMs lead on logic, JSON output and RAG — 100k tokens of manuals is easy in text, painful in an audio KV cache.



Modular & swappable

Need a new voice? Swap only the TTS model. STT and LLM stay untouched; each stage upgrades on its own schedule.



Easy guardrails

Run regex, NeMo Guardrails or Llama Guard on the intermediate text and block unsafe output before it ever reaches TTS.



Full observability

An exact transcript of what the user said and what the AI replied makes debugging, QA and auditing straightforward.

The Trade-offs

What gets lost — and what gets expensive — when speech becomes text.



Paralinguistic blind spot

Text strips tone. A heavy sigh plus “Yeah, sure, whatever” is transcribed as agreement — and answered cheerfully.



The latency tax

Time-to-first-audio is the sum of every stage. Sub-800 ms takes heroic engineering.



Barge-in complexity

Interruptions need a separate VAD, instant TTS cancellation, buffer flushes and race-free state handling.

Where time-to-first-audio goes

Illustrative budget for an ~800 ms target

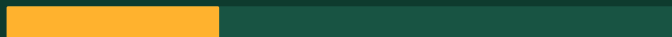
VAD tail-padding 200 ms



STT inference 150 ms



LLM prefill + 1st token 250 ms



TTS prefill + 1st chunk 200 ms



Cascaded vs. Omni (Native Audio)

Feature	Cascaded (STT → LLM → TTS)	Omni (audio ↔ audio)
Latency (TTFA)	~800 ms – 1.5 s; heavy optimisation	~300 – 500 ms; near human
Emotion & tone	Lost in transcription	Preserved and mirrored
Barge-in	External VAD + state machine	Native — model stops generating
Reasoning / RAG	Excellent — mature text LLMs	Poor / unreliable
Guardrails	Easy — filter text before TTS	Hard — must filter audio
Hallucinations	Textual, easy to catch	Acoustic artefacts, odd voices
Voice changes	Swap the TTS model	Baked into weights; fine-tune
Serving	Multi-model routing	Huge audio KV-cache
Cost / minute	Moderate; scale stages independently	High; large unified GPU memory

Green = advantage • Red = limitation

Choosing an Architecture

Cascaded trades latency and emotional nuance for control, accuracy and flexibility.



Cascaded fits when...

- Answers depend on documents, tools or strict JSON
- Compliance needs transcripts and text guardrails
- Brand voices must be swappable



Omni fits when...

- Natural, sub-500 ms conversation is the product
- Tone and emotion must be heard and mirrored
- Users interrupt often; reasoning is modest

EXAMPLE OMNI / SPEECH-TO-SPEECH MODELS

Qwen3-Omni

Alibaba · open

Moshi

Kyutai · open

MiniCPM-o

OpenBMB · open

GLM-4-Voice

Zhipu AI · open

GPT Realtime

OpenAI · API

Gemini Live

Google · API